# POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year: 2025
Programme: BCSIT | Full Marks: 100
Course: Numerical Methods | Pass Marks: 45
Time: 3 hrs.

*Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.*

Section "A"

**Very Short Answer Questions**

Attempt all the questions. [10×2=20]

1. Define truncation error and round-off error with examples.
2. What is the main advantage of Newton-Raphson over the bisection method?
3. What is an ill-conditioned system of linear equations?
4. Why Gauss Seidel method is better than Jacobi's method?
5. Define interpolation. Mention one use of interpolation.
6. Use trapezoidal rule to evaluate $\int_0^1 x^3\,dx$ (n=1)
7. State the formula of Simpson's 1/3 rule.
8. What is the difference between initial value problem (IVP) and boundary value problem (BVP)?
9. Differentiate between the bracketing and non-bracketing method of finding the solution of the nonlinear equations with example.
10. Write the equation for Taylor series.

Section "B"

**Descriptive Answer Questions**

Attempt **any five** questions. [5×10=50]

11. Define false position method. Find the root of the equation $\cos x = e^x$ using false position method correct up to 4 decimal places.
12. Find the largest eigen value and the corresponding eigen vector of matrix.
$$\begin{bmatrix} 1 & -1 & -1 \\ 3 & -1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$$
13. Evaluate: $\int_0^1 \frac{dx}{1+x^2}$ using:
    a. Simpson's 1/3rd rule taking h=1/4.
    b. Simpson's 3/8th rule taking h=1/6.
14. Evaluate $\int_0^{\pi} Sinx\ dx$, using Gauss formula for n=2 and n=3 and compare their values with exact solution.

15. Apply Gauss-Jordan method to solve the following equations:
$x + y + z = 9;\ 2x - 3y + 4z = 13;\ 3x + 4y + 5z = 40.$

16. Fit the exponential curves $y=ae^{bx}$

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| y | 1.05 | 2.10 | 3.85 | 8.30 |

Group "C"

**Long Answer Questions**

Attempt **any two** questions. [2×15=30]

17. a. Solve the Poisson's equation $\nabla^2 f = -10(x^2 + y^2 + 10)$ over the square domain $0 \le x \le 3$, $0 \le y \le 3$ with $f=0$ on the boundary and mesh length $h=1$. Use Gauss-Seidel method to solve the equation up to fifth iteration. [12]

    b. What are the real-life applications of Laplace and Poisson's Equation. [3]

18. a. Find the inverse of matrix using Matrix Inversion method (Gauss Jordon). [8]

$$\begin{bmatrix} 1 & -1 & -1 \\ 3 & -1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$$

    b. Find the square root of 7 correct up to 4 decimal place using N-R (Newton-Raphson) method**.** [7]

19. a. What do you mean by shooting method? [1]

    b. Given the equation
    $dy/dx = y + e^x$ with $y(0) = 0$
    Estimate y(0.2) and y(0.4) using Heun's method. [10]

    c. Write an algorithm for this method for the solution of 1st order differential equations. [4]